

Blockchain and Cryptocurrency in Business: Development, Corporate Adoption, and Impact (2009-2026)

*A Research Brief on Technology Evolution, Corporate Strategy, Market Cycles,
Regulation, Accounting, and Systemic Risk*

10 July 2026

Abstract

This research brief examines the development, corporate adoption, and impact of blockchain and cryptocurrency in business over the period from Bitcoin's emergence in 2009 through mid-2026. Drawing on verified market data, regulatory sources, corporate filings, and a systematic review of academic literature, the brief maps the arc from a peer-to-peer electronic cash experiment to a multi-trillion-dollar asset class with profound implications for corporate finance, accounting, and regulation.

1. Introduction and Motivation

On 31 October 2008, a person or group using the name Satoshi Nakamoto posted a nine-page white paper to a cryptography mailing list. The paper, titled "Bitcoin: A Peer-to-Peer Electronic Cash System," proposed a system of digital payments that did not rely on a trusted intermediary: transactions would be verified by a distributed network of participants through cryptographic proof, rather than by a bank or payment processor (Nakamoto, 2008). On 3 January 2009, the Bitcoin network went live with the mining of the genesis block, embedding the headline from that day's Times of London: "Chancellor on brink of second bailout for banks." The message was both a timestamp and a statement of purpose.

Seventeen years later, the cryptocurrency and blockchain ecosystem has evolved into a multi-trillion-dollar asset class, an active field of academic inquiry, and a regulatory challenge spanning every major jurisdiction. The journey has not been linear. It has been punctuated by spectacular booms and busts, by the rise and fall of exchanges and protocols, by the entry of the world's largest financial institutions into crypto markets, and by the gradual, uneven construction of a regulatory architecture that was nowhere in sight when Nakamoto posted that mailing list message.

This research brief provides a synthetic analysis of blockchain and cryptocurrency in business, with a focus on corporate adoption, accounting and reporting, regulation, and systemic risk. It covers the period from Bitcoin's emergence (2009) through the ICO wave (2017-2018), the DeFi and NFT booms (2020-2021), the cascade of exchange and protocol failures (2022), the institutional pivot through ETFs (2024), and the consolidation of regulatory frameworks in the EU, US, UK, and Asia-Pacific through mid-2026.

The academic motivation for this review is twofold. First, blockchain and cryptocurrency represent a genuinely novel institutional phenomenon: a technology that combines monetary economics, platform economics, corporate governance, financial reporting, and securities regulation in ways that existing academic frameworks struggle to capture fully. The literature has responded with theoretical work on tokenomics (Cong, Li, and Wang, 2021; Sockin and Xiong, 2023), empirical studies of ICO disclosure (Bourveau et al., 2022, JAR; Howell, Niessner, and Yermack, 2020, RFS), and foundational economic appraisals of cryptocurrency pricing and risk (Liu, Tsyvinski, and Wu, 2022, JF; Biais et al., 2023, JF). Second, the regulatory and accounting frameworks that govern crypto assets are evolving rapidly, creating natural experiments that accounting and finance scholars are only beginning to exploit. FASB ASU 2023-08, the EU MiCA regulation, and the approval of spot crypto ETFs in the US all represent structural breaks in the governance of digital assets.

The brief proceeds as follows. Section 2 charts the evolution of blockchain technology and its business use cases. Section 3 analyses market development and cycles. Section 4 examines corporate adoption and treasury exposure. Section 5 reviews accounting, auditing, and reporting issues. Section 6 surveys the global regulatory landscape. Section 7 assesses systemic and firm-

level risks. Section 8 identifies hot themes and emerging areas. Section 9 maps research gaps and opportunities. Section 10 concludes. All data and methodology documentation are freely available for download.

2. Evolution of the Technology and Business Use Cases

Bitcoin's original design was simple in conception: a peer-to-peer payment network secured by proof-of-work mining, with a fixed supply of 21 million coins. In practice, Bitcoin's utility as a medium of exchange was limited from the outset by extreme price volatility and slow transaction settlement. Yermack (2015), in the first rigorous economic appraisal of Bitcoin, concluded that it largely failed to satisfy the criteria of a bona fide currency: it functioned poorly as a medium of exchange, a store of value, and a unit of account. Easley, O'Hara, and Basu (2019, JFE) later formalised Bitcoin's fee market dynamics, showing how limited block space and user willingness to pay for priority settlement created a transaction fee structure that increasingly resembled a congestion pricing mechanism rather than a payments rail.

Over time, Bitcoin's narrative shifted from "peer-to-peer electronic cash" to "digital gold": a scarce asset with no central issuer, protected against inflation by its fixed supply schedule, and increasingly held by institutional investors as a portfolio diversifier. This narrative shift, while controversial, was reinforced by the halving events of 2012, 2016, 2020, and 2024, each of which reduced the block reward by 50 percent, and by the growing body of research that treats Bitcoin as a distinct asset class with its own risk factors (Liu, Tsyvinski, and Wu, 2022, JF).

The Bitcoin network itself has seen limited technological evolution compared to subsequent blockchains. The Taproot upgrade (2021) improved privacy and smart contract flexibility. The Lightning Network, a second-layer protocol enabling faster and cheaper payments, has grown to over 5,000 BTC in channel capacity by mid-2026, but adoption remains concentrated among a small subset of users and routing nodes. SegWit adoption gradually increased transaction throughput, but Bitcoin's base layer processes approximately 7 transactions per second, a constraint that is structural rather than incidental.

Ethereum, proposed by Vitalik Buterin in late 2013 and launched in July 2015, introduced the concept of a Turing-complete blockchain: a distributed computer capable of executing arbitrary code in the form of "smart contracts." This was a qualitative shift from Bitcoin's limited scripting language. Smart contracts enable self-executing agreements where the terms are written in code and automatically enforced by the network, without reliance on a legal system or trusted intermediary.

Cong and He (2019, RFS) provided the first formal economic analysis of blockchain-based smart contracts, showing that while they can mitigate information asymmetry and improve welfare, the very transparency that enables consensus generation may also encourage greater collusion among participants who observe each other's actions. This tension between transparency benefits and collusion risks has become a recurring theme in blockchain analysis.

Ethereum's shift from proof-of-work to proof-of-stake (the "Merge," completed on 15 September 2022) was the largest consensus mechanism migration in blockchain history. Saleh (2021, RFS) had earlier demonstrated that proof-of-stake requires only a modest reward schedule to achieve equilibrium with immediate consensus and no persistent forking, but also that validators must be stakeholders for the mechanism to work. The Merge reduced Ethereum's energy consumption by approximately 99.9 percent, addressing the most prominent criticism of proof-of-work blockchain technology. Capponi, Olafsson, and Alsabah (2023, MS) showed that mining hardware efficiency competition in proof-of-work systems can lead to centralisation, lending theoretical support to the shift to staking-based consensus.

Between 2015 and 2019, a wave of enterprise blockchain initiatives sought to apply distributed ledger technology to business problems without the volatility, regulatory uncertainty, and public-chain constraints associated with open blockchains. The dominant architecture was the permissioned or consortium blockchain, where known participants operate a shared ledger governed by a predefined set of rules. Hyperledger Fabric, R3 Corda, and Quorum were the leading platforms.

The track record of these initiatives is mixed. IBM Food Trust, built on Hyperledger Fabric, remains operational with production deployments at Walmart (where traceability for leafy greens was reduced from 7 days to 2.2 seconds), Carrefour, Nestle, and other retailers. R3 Corda reportedly processed over 1 million transactions per day by 2025, powering live trade finance networks with clients including DTCC, Nasdaq, and HSBC.

In contrast, TradeLens, the IBM-Maersk joint venture for digitising global shipping supply chains, was discontinued in November 2022 after failing to achieve commercial viability. The official statement cited "the need for full global industry collaboration" that "has not been achieved" and the inability to "meet the financial expectations as an independent business" (Maersk, 2022). TradeLens had onboarded over 600 ports and terminals and 175 organisations at its peak, demonstrating that technological functionality alone does not guarantee commercial success: network effects in supply chains require near-universal adoption to deliver value, and the coordination costs of achieving that adoption proved insurmountable in a fragmented shipping industry.

The enterprise blockchain narrative has shifted substantially since 2022. Rather than building bespoke consortium chains, many firms now build on public blockchain infrastructure (particularly Ethereum) with permissioned access controls, or use blockchain-as-a-service (BaaS) offerings from cloud providers. The global blockchain market reached an estimated \$33 billion in 2025, with BaaS from Microsoft Azure, AWS, and Oracle becoming the default entry point for most corporate deployments.

Enterprise blockchain lesson: TradeLens's failure illustrates a broader pattern. Blockchain solutions for inter-firm coordination face a collective action problem: the benefits accrue only when most or all participants adopt the system, but individual firms have limited

incentive to join until others have already joined. Without a dominant industry participant or regulatory mandate to force adoption, many consortium blockchain initiatives stall in the pilot phase. IBM Food Trust succeeded where TradeLens failed partly because Walmart, as a dominant buyer, could mandate supplier participation unilaterally.

Payments and settlement. Cryptocurrency has found its most natural business application in cross-border payments and remittances, where traditional correspondent banking is slow (2 to 5 days), expensive (average cost of 6.4 percent for remittances as of Q4 2024 according to the World Bank), and opaque. Stablecoins, in particular, have become a significant payments infrastructure: Visa reported approximately \$3.5 billion in annualised USDC settlement volume by late 2025, and Stripe's acquisition of stablecoin infrastructure provider Bridge in 2025 signalled that major payment processors view stablecoin rails as a strategic priority. The BIS Innovation Hub has run multiple proof-of-concept projects demonstrating that tokenised wholesale settlement can reduce cross-border payment costs and settlement times substantially.

Trade finance. Blockchain-based trade finance networks address the paper-intensive, multi-party, trust-dependent nature of international trade. R3 Corda's Marco Polo and we.trade networks (the latter discontinued in 2022) demonstrated the technology's potential, but the same coordination problems that affected TradeLens have limited the scale of deployment in trade finance. The most successful initiatives are those with strong institutional backing, such as Contour (backed by major banks including HSBC, Standard Chartered, and BNP Paribas), which digitises letters of credit on Corda.

Tokenisation of real-world assets. The representation of traditional financial assets (bonds, equities, real estate, commodities) as tokens on a blockchain has attracted substantial institutional interest. BlackRock's BUIDL fund, launched in March 2024 on Ethereum, tokenises a money market fund and had accumulated over \$500 million in assets by late 2024. The Boston Consulting Group has estimated that the market for tokenised assets could reach \$16 trillion by 2030, a projection that should be treated cautiously given the industry's history of overoptimistic forecasts.

Identity and credentials. Self-sovereign identity systems built on blockchain aim to give individuals control over their digital identities and credentials, reducing reliance on centralised identity providers. The EU's European Blockchain Services Infrastructure (EBSI) is developing cross-border digital identity solutions using blockchain. Practical deployment remains at an early stage, with the World Bank's Mission Billion initiative and the Sovrin Foundation among the leading implementations.

2.1 Bitcoin and the First Generation: Digital Cash and Store of Value

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2.4 Business Use Cases: Payments, Trade Finance, Tokenisation, and Identity

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3. Market Development and Cycles

The cryptocurrency market has exhibited a recurring boom-and-bust pattern over its 15-year history, with each cycle producing a new all-time high followed by a drawdown of 70 to 90 percent. Pagnotta (2022, RFS) provides a theoretical framework for understanding this volatility, showing that Bitcoin's design leads to multiple equilibria: the feedback between price and network security can amplify fundamental shocks and produce booms and busts disconnected

from fundamentals. Biais et al. (2023, JF) formalise this insight in a general equilibrium model where equilibrium prices reflect sunspots, creating multiple equilibria and extrinsic volatility.

The first major cycle occurred in 2013-2014: Bitcoin rose from approximately \$13 to over \$1,100 in November 2013, driven by early adopters and the first wave of media attention, then collapsed to approximately \$200 following the Mt. Gox exchange failure in February 2014. The second cycle (2017-2018) was driven by the ICO boom: Bitcoin reached nearly \$20,000 and the total market cap approached \$800 billion before falling 85 percent to a low of \$3,200 by December 2018. The third cycle (2020-2022) reflected the convergence of DeFi, NFTs, and institutional entry, with the market peaking at \$3 trillion in November 2021 before the Terra/LUNA and FTX collapses drove valuations back to \$800 billion by year-end 2022.

The fourth cycle, beginning in 2023 and continuing through mid-2026, was distinguished by two features absent from earlier cycles: institutional infrastructure (spot ETFs, custody by systemically important banks, regulatory frameworks), and the decoupling of Bitcoin narrative from the broader crypto market. Bitcoin rebounded from its 2022 low of approximately \$16,000 to cross \$100,000 for the first time in December 2024 and reached an all-time high of over \$126,000 in October 2025, while many altcoins traded well below their 2021 peaks.

The year 2022 was the most destructive in the short history of cryptocurrency markets. The cascade began in May 2022 with the collapse of the Terra/LUNA ecosystem: the algorithmic stablecoin TerraUSD (UST) lost its dollar peg, triggering a death spiral that destroyed approximately \$55 billion in market capitalisation within days. The collapse reverberated through the crypto lending sector. Three Arrows Capital (3AC), a Singapore-based hedge fund with approximately \$10 billion in assets under management at its peak, failed to honour margin calls and was forced into liquidation in June 2022, owing creditors over \$3.5 billion. Celsius Network, Voyager Digital, and BlockFi all filed for bankruptcy within weeks, each citing exposure to 3AC, LUNA/UST, or both.

In November 2022, FTX, then the world's second-largest cryptocurrency exchange by volume and valued at \$32 billion at its peak, collapsed over a period of approximately 72 hours. A CoinDesk report on 2 November 2022 revealing that FTX's sister trading firm Alameda Research held a disproportionate share of its assets in FTX's native FTT token triggered a run on deposits. Approximately \$6 billion in customer withdrawals were demanded over three days; FTX could not honour them. The company filed for bankruptcy on 11 November 2022. CEO Sam Bankman-Fried was later convicted on seven counts of fraud and sentenced to 25 years in prison; he was ordered to pay \$11 billion in forfeiture. As of early 2026, liquidators had recovered approximately \$7.3 billion in assets for creditors.

The 2022 failures shared a common feature: they were failures of centralised intermediaries, not of blockchain protocols. The blockchains themselves (Bitcoin, Ethereum, Solana) continued to process transactions throughout the crisis. In this sense, 2022 validated the technological resilience of distributed ledgers while exposing the governance and risk management failures of the centralised institutions that had grown up around them. Griffin and Shams (2020, JF) had

earlier provided evidence that Tether, a centralised stablecoin issuer, systematically influenced Bitcoin prices during the 2017 boom through purchases timed to follow market downturns, a finding that anticipated the centralisation-related vulnerabilities that would materialise in 2022.

The integrity of cryptocurrency trading volumes has been a persistent concern. Cong et al. (2023, MS) provided the first systematic methodology for detecting wash trading on crypto exchanges and found that unregulated exchanges display wash trading averaging over 70 percent of reported volume, fabricating trillions of dollars annually to improve exchange rankings and temporarily distort prices. Their finding implies that widely cited trading volume statistics systematically overstate genuine market activity. Makarov and Schoar (2020, JFE) documented large, recurrent arbitrage opportunities across cryptocurrency exchanges, with price deviations larger across countries (due to capital controls) than between cryptocurrencies. These findings collectively suggest that cryptocurrency markets, despite their technological sophistication, remain substantially less integrated and less transparent than traditional equity markets.

The approval of 11 spot Bitcoin ETFs by the US SEC on 10 January 2024, followed by 8 spot Ether ETFs on 23 May 2024, represented the largest structural shift in cryptocurrency market access since the launch of Bitcoin futures in December 2017. Augustin, Rubtsov, and Shin (2023, MS) had earlier shown that the introduction of Bitcoin futures improved spot market quality by reducing volatility and increasing liquidity and informational efficiency. The spot ETF approvals took this effect further by enabling regulated, exchange-traded access to crypto exposure for the entire universe of brokerage accounts, retirement plans, and institutional portfolios.

As of 9 July 2026, spot Bitcoin ETFs had accumulated \$51.3 billion in cumulative net inflows. BlackRock's iShares Bitcoin Trust (IBIT) alone accounts for \$60.2 billion of gross inflows, partially offset by the persistent drain from the legacy Grayscale Bitcoin Trust (GBTC, \$27.3 billion in outflows). Spot Ether ETFs reached \$11.0 billion in cumulative net inflows, with the same pattern of BlackRock dominance and Grayscale legacy outflow. Total ETH ETF assets represented 4.57 percent of Ether's circulating market capitalisation as of June 2026 (CoinDesk).

3.1 Boom-and-Bust Cycles: A Persistent Pattern

The cryptocurrency market has exhibited a recurring boom-and-bust pattern over its 15-year history, with each cycle producing a new all-time high followed by a drawdown of 70 to 90 percent. Pagnotta (2022, RFS) provides a theoretical framework for understanding this volatility, showing that Bitcoin's design leads to multiple equilibria: the feedback between price and network security can amplify fundamental shocks and produce booms and busts disconnected from fundamentals. Biais et al. (2023, JF) formalise this insight in a general equilibrium model where equilibrium prices reflect sunspots, creating multiple equilibria and extrinsic volatility.

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3.3 Wash Trading and Market Integrity

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3.4 The ETF Era: Institutional Infrastructure

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4. Corporate Adoption and Treasury Exposure

The most consequential corporate adoption story in the history of cryptocurrency is Strategy (formerly MicroStrategy, ticker MSTR). On 11 August 2020, the business intelligence software company announced its first Bitcoin purchase: 21,454 BTC for \$250 million, at an average price of approximately \$11,653 per coin. In a conference call the same month, CEO Michael Saylor described Bitcoin as "a dependable store of value and an attractive investment asset with more long-term appreciation potential than holding cash."

What followed was unprecedented. Over the next six years, Strategy accumulated Bitcoin through a combination of cash from operations, convertible note issuances, and at-the-market equity offerings. As of 6 July 2026, the company holds 843,775 BTC at a cumulative cost basis of \$63.7 billion (average \$75,476 per BTC). The holding represents approximately 4 percent of the total Bitcoin supply. The company files an 8-K with the SEC disclosing every purchase, typically within days, creating the most transparent corporate treasury programme in history.

The economic logic of Strategy's Bitcoin strategy has been subject to academic scrutiny and market debate. Proponents argue that Bitcoin serves as an inflation hedge and a superior store of value relative to cash equivalents yielding negative real returns in accommodative monetary environments. Critics note that a software company converting substantially all of its treasury into a volatile digital asset represents a fundamental change in the firm's business model, substituting Bitcoin price exposure for operational revenue generation as the primary driver of equity value. The market's verdict has been extreme: MSTR shares traded at approximately \$120 before the first Bitcoin purchase in August 2020 and exceeded \$2,000 by early 2025 (split-adjusted). The company changed its name to Strategy in February 2025, formally recognising that Bitcoin treasury management had become its defining business activity.

Beyond Strategy, the corporate Bitcoin treasury landscape is concentrated in crypto-native firms (miners, exchanges) and a small number of technology companies. The motivations vary: miners hold a portion of production as treasury assets; exchanges hold customer and proprietary assets; and a few non-crypto firms have made strategic allocations.

Tesla purchased \$1.5 billion in Bitcoin in February 2021 and briefly accepted it as payment for vehicles before suspending the programme in May 2021, citing environmental concerns. The company sold 75 percent of its position in July 2022 at a loss, retaining 11,509 BTC valued at \$386 million (\$33,539 per BTC) as of its December 2024 10-K. Block (formerly Square) holds 9,032 BTC and generated \$24.1 billion in Bitcoin revenue through its Cash App in 2025, highlighting the distinction between treasury holdings and transaction-facilitation revenue. Among crypto-native firms, Marathon Digital Holdings holds the largest treasury at approximately 38,689 BTC (March 2026), followed by Galaxy Digital (25,723 BTC), Coinbase (16,492 BTC), Riot Platforms (15,679 BTC), and CleanSpark (13,924 BTC).

By January 2026, 60 percent of the top 25 US banks (15 of 25) offered or were developing Bitcoin-related services including trading, custody, lending, and advisory (River Research, cited by [earnpark.com](#)). BNY Mellon, the first global systemically important bank to custody crypto assets (October 2022), received SEC non-objection in September 2024 for a custody structure using individual crypto wallets without balance-sheet liability recognition, a regulatory innovation that substantially lowered the capital cost of providing custody services. The SEC's rescission of SAB 121 (which had required banks to record custodied crypto as liabilities) further opened institutional custody markets in 2025.

The involvement of traditional financial institutions in crypto markets has accelerated since 2024. The approval of spot ETFs created a new, regulated distribution channel: Morgan Stanley,

Wells Fargo, and Bank of America now offer Bitcoin ETF access to wealth management clients. JPMorgan, which CEO Jamie Dimon famously called Bitcoin a "fraud" in 2017, has been exploring institutional crypto trading since 2025. Goldman Sachs provides Bitcoin exposure via ETFs, operates a private blockchain testing tokenised fund redemptions, and advises high-net-worth clients on crypto allocations.

Payment providers have moved further and faster than banks. PayPal launched its PYUSD stablecoin (issued by Paxos Trust Company) in August 2023, offering it on PayPal and Venmo with 4 percent annual rewards on holdings as of mid-2026. Visa enabled USDC settlement for selected issuers in 2023 and expanded to approximately \$3.5 billion in annualised settlement volume across Ethereum and Solana by late 2025. Mastercard joined the USDG stablecoin consortium in June 2025 and added support for PYUSD, FIUSD, and USDC on its Multi-Token Network. Stripe re-entered crypto infrastructure in 2022 and acquired Bridge, a stablecoin infrastructure provider, in 2025. These developments collectively signal that major payment rails view stablecoins as a strategic complement to, and potential replacement for, legacy correspondent banking infrastructure in cross-border payments.

Table 1: Major Payment Provider and Bank Crypto Services (2023-2026)

Institution	Category	Key Crypto Service	Status (mid-2026)
BNY Mellon	Bank/Custody	Bitcoin/Ether custody; tokenised deposits	Active (Oct 2022 launch)
JPMorgan	Bank/Trading	Institutional crypto trading (developing)	Developing
Goldman Sachs	Bank/Advisory	ETF access; tokenised fund redemptions	Active
Morgan Stanley	Bank/Wealth	Bitcoin ETF access; E*Trade BTC trading (H1 2026)	Active
PayPal	Payment	PYUSD stablecoin; 4% yield on holdings	Active (Aug 2023 launch)
Visa	Payment	USDC settlement (~\$3.5B annualised)	Active
Mastercard	Payment	Multi-Token Network; USDG consortium; stablecoin support	Active
Stripe	Payment	Bridge acquisition; stablecoin-native payments	Active
Fiserv	Payment	FIUSD stablecoin; Mastercard Multi-Token Network	Active

4.1 Bitcoin on the Balance Sheet: Strategy and the Corporate Treasury Movement

The most consequential corporate adoption story in the history of cryptocurrency is Strategy (formerly MicroStrategy, ticker MSTR). On 11 August 2020, the business intelligence software company announced its first Bitcoin purchase: 21,454 BTC for \$250 million, at an average price of approximately \$11,653 per coin. In a conference call the same month, CEO Michael Saylor described Bitcoin as "a dependable store of value and an attractive investment asset with more long-term appreciation potential than holding cash."

What followed was unprecedented. Over the next six years, Strategy accumulated Bitcoin through a combination of cash from operations, convertible note issuances, and at-the-market equity offerings. As of 6 July 2026, the company holds 843,775 BTC at a cumulative cost basis of \$63.7 billion (average \$75,476 per BTC). The holding represents approximately 4 percent of the total Bitcoin supply. The company files an 8-K with the SEC disclosing every purchase, typically within days, creating the most transparent corporate treasury programme in history.

The economic logic of Strategy's Bitcoin strategy has been subject to academic scrutiny and market debate. Proponents argue that Bitcoin serves as an inflation hedge and a superior store of value relative to cash equivalents yielding negative real returns in accommodative monetary environments. Critics note that a software company converting substantially all of its treasury into a volatile digital asset represents a fundamental change in the firm's business model, substituting Bitcoin price exposure for operational revenue generation as the primary driver of equity value. The market's verdict has been extreme: MSTR shares traded at approximately \$120 before the first Bitcoin purchase in August 2020 and exceeded \$2,000 by early 2025 (split-adjusted). The company changed its name to Strategy in February 2025, formally recognising that Bitcoin treasury management had become its defining business activity.

4.2 Other Corporate Holders: Motivations and Scale

Beyond Strategy, the corporate Bitcoin treasury landscape is concentrated in crypto-native firms (miners, exchanges) and a small number of technology companies. The motivations vary: miners hold a portion of production as treasury assets; exchanges hold customer and proprietary assets; and a few non-crypto firms have made strategic allocations.

Tesla purchased \$1.5 billion in Bitcoin in February 2021 and briefly accepted it as payment for vehicles before suspending the programme in May 2021, citing environmental concerns. The company sold 75 percent of its position in July 2022 at a loss, retaining 11,509 BTC valued at \$386 million (\$33,539 per BTC) as of its December 2024 10-K. Block (formerly Square) holds 9,032 BTC and generated \$24.1 billion in Bitcoin revenue through its Cash App in 2025, highlighting the distinction between treasury holdings and transaction-facilitation revenue. Among crypto-native firms, Marathon Digital Holdings holds the largest treasury at approximately 38,689 BTC (March 2026), followed by Galaxy Digital (25,723 BTC), Coinbase (16,492 BTC), Riot Platforms (15,679 BTC), and CleanSpark (13,924 BTC).

By January 2026, 60 percent of the top 25 US banks (15 of 25) offered or were developing Bitcoin-related services including trading, custody, lending, and advisory (River Research, cited by earnpark.com). BNY Mellon, the first global systemically important bank to custody crypto assets (October 2022), received SEC non-objection in September 2024 for a custody structure using individual crypto wallets without balance-sheet liability recognition, a regulatory innovation that substantially lowered the capital cost of providing custody services. The SEC's rescission of SAB 121 (which had required banks to record custodied crypto as liabilities) further opened institutional custody markets in 2025.

4.3 Bank and Payment Provider Involvement

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5. Accounting, Auditing, and Reporting Issues

For the first decade of corporate crypto exposure, US GAAP provided no specific guidance for digital assets. The prevailing treatment, confirmed by the AICPA and large audit firms, treated crypto holdings as indefinite-lived intangible assets under ASC 350. This classification created a structural accounting asymmetry: crypto assets were recorded at cost, tested for impairment whenever their fair value fell below carrying value, but never revalued upward when prices recovered. A firm holding Bitcoin at a cost basis of \$30,000 would record an impairment charge if the price fell to \$20,000, but could not reverse that charge when the price rose to \$60,000. The asset remained on the balance sheet at \$20,000 indefinitely, unless the firm sold it, at which point a gain would be recognised in the income statement.

This treatment was widely criticised by preparers, auditors, and analysts. It distorted the balance sheet by systematically understating asset values during bull markets. It created perverse incentives: firms that adopted early and held through a cycle would report large impairment losses followed by permanently depressed carrying values, while firms that entered at a market trough avoided comparable accounting penalties. Luo and Yu (2024, RAST) documented significant inconsistencies in measurement bases, classification, and cash flow reporting across 40 global firms with crypto exposure, finding that these inconsistencies could mislead financial statement users.

Anderson et al. (2026, JAR) provide the most comprehensive analysis of US public firms' cryptocurrency holdings and accounting practices from 2013 to 2022. They find that firms view crypto assets "more akin to investments rather than intangible assets," a finding that anticipated the FASB's eventual fair-value determination. The paper documents substantial heterogeneity in both the level of holdings and the accounting choices firms made in the absence of explicit guidance, including variation in impairment trigger definitions, unit-of-account determinations, and disclosure practices.

On 13 December 2023, the FASB issued ASU 2023-08, "Intangibles -- Goodwill and Other -- Crypto Assets" (ASC Subtopic 350-60). The standard requires in-scope crypto assets to be

measured at fair value under ASC 820, with changes in fair value recognised in net income each reporting period. The scope is limited to assets that meet the definition of an intangible asset, reside on a blockchain or distributed ledger, are secured by cryptography, are fungible, and are not created or issued by the reporting entity or its related parties. The standard also requires extensive disclosures: name, cost basis, fair value, and units held for each significant crypto asset holding, as well as aggregate information for non-significant holdings.

The standard is effective for fiscal years beginning after 15 December 2024, including interim periods, with early adoption permitted. This means that the first wave of fair-value crypto financial statements is being filed in 2025 and 2026 for calendar-year public companies.

The accounting effect is substantial. Under the impairment model, a firm holding Bitcoin with a cost basis of \$30,000 during the 2022 bear market (when Bitcoin fell below \$17,000) would have reported an impairment loss of over \$13,000 per coin. Under fair value, the same firm would report unrealised gains or losses each period based on market prices, with the asset on the balance sheet reflecting current fair value. The standard eliminates the asymmetric treatment that had been the single largest accounting deterrent to corporate crypto holdings. It also introduces earnings volatility: quarterly net income will now reflect the quarter's Bitcoin price movement, a consideration that boards and audit committees must weigh against the informational benefit of faithful representation.

Under IFRS, the treatment of cryptocurrency holdings remains governed by the IFRS Interpretations Committee's June 2019 Agenda Decision, which concluded that crypto should be accounted for either as intangible assets under IAS 38 (at cost, with optional revaluation to fair value if an active market exists) or as inventory under IAS 2 for broker-traders. The IAS 38 revaluation option nominally allows fair value measurement, but it requires that revaluation gains be recognised in other comprehensive income (not profit or loss) and that the fair value be reliably measurable (a higher threshold than ASC 820). In practice, few IFRS-reporting entities use the revaluation model for crypto assets.

The IASB has not yet added a dedicated crypto project to its workplan, despite requests from stakeholders during the 2021 Agenda Consultation. The Board's ongoing comprehensive review of IAS 38 may eventually address cryptocurrency treatment, but no new standard is expected in the near term. The divergence between US GAAP (fair value through net income) and IFRS (cost with optional OCI revaluation) creates material differences in reported financial positions for firms with crypto exposure reporting under the two frameworks, an area that comparative accounting researchers are only beginning to explore.

Research opportunity: The FASB's adoption of fair value measurement for crypto assets creates a natural experiment. Researchers can compare the earnings and balance sheet effects for firms that adopted early versus those that adopted at the mandatory date, examine whether fair value measurement changes the market's pricing of crypto holdings, and

investigate whether the standard affects corporate treasury allocation decisions. The IFRS-GAAP divergence adds a cross-jurisdictional dimension to these research questions.

The auditability of on-chain data presents both opportunities and challenges. On one hand, public blockchains provide a transparent, immutable, and real-time record of transactions that, in principle, could support continuous auditing. On the other hand, proving that a set of blockchain addresses belongs to a specific entity, verifying that private keys are controlled (not just owned) by the entity, and reconciling on-chain balances with off-chain obligations (such as customer deposits) remain significant audit challenges.

The FTX collapse illustrated these challenges in dramatic form. The exchange's balance sheet, as reflected in bankruptcy filings, showed a fundamental mismatch between on-chain assets and customer obligations that was not detected by the firm's auditors. In response, the concept of "proof of reserves" (PoR) gained traction: a cryptographic attestation that an exchange or custodian holds sufficient on-chain assets to cover customer deposits, verified by an independent third party. Bourveau, Brendel, and Schoenfeld (2024, RAST) provide the first evidence on smart contract audit reports in DeFi, analysing approximately 8,500 audit reports and showing that these audits are pervasive, that the market consists of new technical audit firms distinct from traditional accounting firms, and that the market reacts positively to audit report releases.

However, PoR has limitations. It verifies assets but not liabilities: an exchange can prove it holds 100,000 BTC on-chain without disclosing that customer obligations total 150,000 BTC. It does not address the going-concern risk of the entity. And the technical standards for PoR engagements are not established by any accounting standard-setter, meaning that the quality and rigour of PoR reports vary substantially across providers. The PCAOB and IAASB have not yet issued guidance on blockchain attestation engagements, though both have included digital assets in their research agendas.

5.1 The Pre-ASU 2023-08 Regime: Impairment Without Recovery

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5.2 FASB ASU 2023-08: Fair Value Measurement

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5.3 IFRS Treatment and the IASB Gap

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5.4 Auditing and Proof of Reserves

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6. Global Regulatory Landscape and Its Divergence

The EU's Markets in Crypto-Assets Regulation (MiCA), adopted by the European Parliament on 20 April 2023 and published in the Official Journal on 9 June 2023, represents the first comprehensive, cross-border regulatory framework for crypto-assets. MiCA entered into force on 29 June 2023, with Title III and IV (asset-referenced tokens and e-money tokens) becoming applicable on 30 June 2024, and the full framework (including crypto-asset service providers, or CASPs) becoming applicable on 30 December 2024.

MiCA establishes a licensing regime for CASPs, requiring them to meet organisational, governance, and capital requirements. It imposes disclosure obligations on issuers of crypto-assets, including a white paper requirement modelled on securities prospectuses but adapted for the technical characteristics of blockchain-based instruments. It creates specific regimes for asset-referenced tokens (stablecoins backed by a basket of assets) and e-money tokens (stablecoins referencing a single fiat currency), with enhanced prudential requirements for "significant" tokens that reach systemic scale. And it includes market abuse provisions covering insider trading and market manipulation in crypto-assets admitted to trading on EU platforms.

MiCA's significance extends beyond its substantive provisions. By creating a single regulatory passport for crypto services across the EU's 27 member states, it addresses the fragmentation that had forced crypto firms to navigate 27 distinct national regimes. By establishing that crypto-assets are a legitimate subject of financial regulation, it provides regulatory certainty that had been absent since Bitcoin's launch. And by taking effect before equivalent frameworks in the US and most other major jurisdictions, it positions the EU as the first mover in crypto regulation, potentially shaping the standards that other jurisdictions will eventually adopt.

The US regulatory approach to crypto has been characterised by fragmentation across multiple federal and state agencies, jurisdictional disputes, and a reliance on enforcement actions rather than comprehensive legislation. The SEC, under Chair Gary Gensler (2021-2025), pursued an aggressive enforcement strategy, bringing actions against major exchanges including Coinbase, Binance, and Kraken on the theory that most crypto tokens are securities subject to SEC registration requirements. The CFTC asserted jurisdiction over Bitcoin and Ether as commodities, creating overlapping and sometimes conflicting regulatory claims.

The SEC's approval of spot Bitcoin ETFs in January 2024 and spot Ether ETFs in May 2024 represented a de facto acknowledgment that Bitcoin and Ether are not securities (ETFs can only hold securities or commodities; the SEC approved these under the 1933 Securities Act rules

applicable to commodity-based trust shares). This created a curious regulatory architecture: the two largest crypto assets by market capitalisation were implicitly classified as commodities suitable for regulated exchange-traded products, while thousands of smaller tokens remained in regulatory limbo, subject to potential SEC enforcement.

The US Senate passed the GENIUS Act (stablecoin regulation) in July 2025, and multiple comprehensive crypto market structure bills have been introduced in Congress. The rescission of SAB 121 (which had required banks to record custodied crypto as liabilities) removed a significant barrier to institutional custody. However, as of mid-2026, the US lacks a comprehensive federal regulatory framework for crypto-assets comparable to MiCA, leaving a patchwork of state-level regulation (such as New York's BitLicense), federal agency guidance, and enforcement actions.

Central bank digital currencies represent the public sector's most direct response to the emergence of private digital money. As of July 2026, 146 countries and currency unions, representing over 98 percent of global GDP, are exploring CBDCs (Atlantic Council CBDC Tracker). Five CBDCs have been launched: the Bahamas Sand Dollar (2020), Nigeria's eNaira (2021), Jamaica's JAM-DEX (2022), and two others. China's e-CNY is by far the largest pilot, having processed over 3.4 billion transactions worth approximately RMB 16.7 trillion (about \$2.3 trillion) by December 2025.

The trajectory of CBDC development reveals an important shift. Early CBDC projects focused on retail applications: digital cash for consumers. More recent activity concentrates on wholesale CBDCs: tokenised central bank money for interbank settlement, cross-border payments, and the settlement of tokenised securities. Thirteen cross-border wholesale CBDC projects are active as of mid-2026, including mBridge (which saw transaction volume surge to \$55.49 billion, a 2,500-fold increase since early-2022 pilots), the ECB's Project Pontes, and Singapore's live wholesale CBDC issuance. Chiu and Davoodalhosseini (2023, MS) provide theoretical support for this trajectory, showing that a cash-like CBDC design (non-interest-bearing, with holding limits) can deliver macroeconomic benefits with limited disruption to bank intermediation.

Brazil's Drex tokenised credit pilot, which runs on a permissioned Ethereum-compatible blockchain, represents a particularly interesting institutional design: a wholesale CBDC that supports the tokenisation of bank deposits and financial assets within a regulated ecosystem. The IMF has praised Drex for its role in driving financial innovation, though it also warned that risks around data privacy, cybersecurity, and financial stability require careful management.

The FATF Travel Rule, extended to virtual assets in June 2019, requires virtual asset service providers (VASPs) to collect and transmit originator and beneficiary information for crypto transfers. Implementation has been fragmented: as of June 2025, 73 percent of jurisdictions (85 of 117 non-prohibiting) had passed Travel Rule legislation, up from 70 percent in 2024 (FATF, 2025). The EU's Transfer of Funds Regulation, complementary to MiCA, took effect on 30 December 2024 with a zero threshold for crypto transfers, the most stringent implementation globally.

Tax treatment of crypto assets varies substantially across jurisdictions. The United States classifies crypto as property, with each disposal (including crypto-to-crypto trades) constituting a taxable event. Japan taxes crypto gains as miscellaneous income at progressive rates up to 55 percent, substantially higher than the 20 percent capital gains rate on listed securities, creating a structural disincentive for domestic crypto investment. Germany exempts crypto gains from tax if the asset is held for more than one year (for private investors), a treatment that encourages long-term holding. Singapore and Hong Kong impose no capital gains tax, though business profits from trading are taxable as income. The UAE has no personal income or capital gains tax, making it an attractive jurisdiction for crypto entrepreneurs and investors.

The divergence in tax treatment creates opportunities for jurisdictional arbitrage, a phenomenon that has attracted the attention of both tax authorities (the OECD's Crypto-Asset Reporting Framework, published in 2022, established a global standard for automatic exchange of crypto tax information) and academic researchers examining the effect of tax policy on crypto market activity and capital flows.

6.1 EU MiCA: The First Comprehensive Framework

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6.2 United States: Fragmentation and the ETF Pivot

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6.3 CBDCs: The Public-Sector Response

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6.4 AML/KYC, Tax, and the Travel Rule

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7. Systemic and Firm-Level Risks

Bitcoin's annualised volatility has averaged approximately 70 to 80 percent over its history, compared to approximately 15 to 20 percent for the S&P 500. For a corporation holding Bitcoin on its balance sheet, this creates direct exposure to earnings volatility under ASU 2023-08's fair value regime. Liu and Tsyvinski (2021, RFS) established that cryptocurrency returns can be predicted by factors capturing momentum and investor attention, but the magnitude of unpredictable variation remains extremely high. Hinzen, John, and Saleh (2022, JFE) demonstrated theoretically that Bitcoin's limited adoption arises as an equilibrium outcome driven by negative network effects: as the network expands, delay grows, settlement becomes

more prolonged, and users abandon the system, creating structural limits to Bitcoin's utility that contribute to its price volatility.

For corporate treasurers, the challenge is not merely the level of volatility but its timing. Crypto drawdowns have historically coincided with broader risk-off episodes in financial markets, reducing Bitcoin's diversification benefit precisely when it is most needed. During the COVID-19 market crash in March 2020, Bitcoin fell approximately 50 percent in a single day, underperforming gold (which fell approximately 12 percent) and US Treasuries (which rallied). This correlation with risk assets, rather than safe havens, complicates the "digital gold" narrative and raises questions about whether crypto exposure adds genuine diversification benefit to a corporate treasury portfolio.

The security of crypto assets depends fundamentally on the security of private key management. For corporations, the custody solution must balance security against operational accessibility: multi-signature wallets, hardware security modules, and institutional-grade custody providers (such as Coinbase Custody, BitGo, and BNY Mellon) offer varying trade-offs. The SEC's rescission of SAB 121 in 2025 removed a significant regulatory disincentive for bank custody of crypto assets, but the operational risk of managing crypto private keys at institutional scale remains an evolving area of practice.

The 2022 cascade of failures illustrated that the most significant risks are not technological (private key theft or blockchain-level attacks) but governance-related (commingling of customer and proprietary funds, inadequate internal controls, related-party transactions). FTX's collapse was not a blockchain failure; it was a failure of corporate governance, internal control, and independent oversight. For accounting and auditing scholars, the 2022 failures provide a rich empirical setting for examining the interaction between external assurance, internal governance, and firm outcomes in an environment with limited regulatory infrastructure.

The association between cryptocurrency and illicit activity has been a persistent reputational and regulatory concern. Foley, Karlsen, and Putnins (2019, RFS) provided the first rigorous empirical estimate, finding that approximately one-quarter of Bitcoin users and one-half of Bitcoin transactions were associated with illegal activity as of 2017, representing approximately \$76 billion in annual illicit transaction volume. Subsequent research and industry data suggest that the proportion of illicit activity has declined as the market has grown: Chainalysis estimated that illicit activity represented 0.34 percent of total crypto transaction volume in 2023, down from 1.9 percent in 2019. Whether this decline reflects genuine improvement or improved laundering techniques that evade detection remains an open question.

Ransomware has become the most economically significant form of crypto-enabled crime. The 2025 Verizon DBIR reports that ransomware was present in 44 percent of all analysed breaches, a 37 percent increase from the prior year. Chainalysis data indicate total ransomware payments peaked at approximately \$1.25 billion in 2023 before declining to \$813 million in 2024 and \$680 million in 2025, driven partly by increasing victim refusal to pay (64 percent refusal rate in 2024, up from approximately 50 percent in 2022) and improved law enforcement disruption.

Bitcoin's proof-of-work consensus mechanism consumes substantial electricity. The Cambridge Bitcoin Electricity Consumption Index estimated annualised consumption of approximately 150 to 170 TWh as of mid-2026, comparable to the electricity consumption of Poland or Egypt. This energy profile has made Bitcoin a target of ESG-focused investors and regulators. Tesla's May 2021 suspension of Bitcoin payments, citing environmental concerns, was a watershed moment in the corporate ESG narrative around crypto.

Several factors complicate the energy narrative. Bitcoin mining is location-agnostic and has increasingly migrated to regions with stranded or excess energy (such as hydroelectric power in Sichuan, China, before the 2021 mining ban, and flared natural gas in Texas). Estimates of Bitcoin's renewable energy share vary widely: the Bitcoin Mining Council, an industry group, claimed 58.9 percent sustainable energy in its Q4 2023 survey; independent academic estimates are typically lower. The Ethereum Merge (September 2022) eliminated approximately 99.9 percent of that network's energy consumption by transitioning to proof-of-stake, demonstrating that the energy problem is specific to proof-of-work, not blockchain technology per se. Capponi, Olafsson, and Alsabah (2023, MS) showed that the same technological arms race that drives energy consumption in proof-of-work systems also undermines their decentralisation goals, providing both an environmental and a governance argument for proof-of-stake alternatives.

A central concern for financial regulators has been whether distress in crypto markets could transmit to the traditional financial system. The 2022 cascade provided a stress test. Despite the destruction of over \$100 billion in market value, the collapse of a \$32 billion exchange, and the failure of multiple crypto lenders, there was no material contagion to traditional financial institutions. Banks with crypto exposure (such as Silvergate and Signature, which served crypto clients) failed due to deposit runs, but the failures were contained within the regional banking sector and did not propagate systemically.

This does not mean that contagion risk is absent. As traditional financial institutions deepen their involvement in crypto markets (through ETF market-making, custody, lending against crypto collateral, and stablecoin integration into payment rails), the channels for potential contagion multiply. The Financial Stability Board (FSB) and the IMF have both identified crypto-asset markets as an area requiring enhanced monitoring and have recommended that jurisdictions apply the principle of "same activity, same risk, same regulation" to crypto activities that are economically equivalent to traditional financial services. The Basel Committee on Banking Supervision has issued prudential standards for bank exposures to crypto-assets, including a 1,250 percent risk weight for unbacked crypto-assets (such as Bitcoin), effectively requiring banks to hold capital equal to the full exposure amount.

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8. Hot Themes and Emerging Areas

The representation of traditional financial assets as blockchain tokens has moved from concept to early-stage implementation. BlackRock's BUIDL fund, JPMorgan's Onyx platform, and the European Investment Bank's digital bond issuances (2021-2023) have demonstrated that regulated financial institutions can issue, trade, and settle tokenised securities using blockchain infrastructure. The Hong Kong Monetary Authority's Project Evergreen and the Monetary Authority of Singapore's Project Guardian have explored tokenised cross-border settlement, establishing legal and technical frameworks that could support wider adoption.

The economic case for tokenisation rests on several potential benefits: fractional ownership of high-value assets (real estate, art, private equity), programmability through smart contracts (automated dividend payments, corporate actions), 24/7 settlement (reducing counterparty risk and collateral requirements), and the composability of tokenised assets within DeFi protocols (using tokenised bonds as collateral for lending). Whether these benefits are sufficient to overcome the coordination costs of shifting financial market infrastructure to new rails is the central unanswered question. Industry projections are optimistic (the Boston Consulting Group estimates a \$16 trillion market by 2030), but the same was said of enterprise blockchain in 2017, before TradeLens, we.trade, and several other high-profile initiatives were discontinued.

Stablecoins have emerged as the most commercially successful application of blockchain technology. Total stablecoin market capitalisation reached \$312 billion as of July 2026 (DeFiLlama), up from approximately \$5 billion in 2020. USDT (Tether) dominates at \$184 billion (59 percent share), followed by USDC at \$73 billion. PayPal's PYUSD, launched in August 2023, reached \$2.85 billion. These are not speculative assets; they are payment instruments pegged to fiat currencies, used for settlement, remittances, and as trading collateral on crypto exchanges.

The integration of stablecoins into traditional payment rails is accelerating. Visa's USDC settlement programme, Mastercard's Multi-Token Network, and Stripe's Bridge acquisition all signal that payment processors view stablecoins as a strategic priority. In cross-border payments,

where traditional correspondent banking is slow and expensive, stablecoins offer near-instant settlement at a fraction of the cost. The BIS Innovation Hub has demonstrated that wholesale CBDCs combined with stablecoin-like tokenised deposits could reduce cross-border payment costs by 50 percent or more.

The regulatory treatment of stablecoins is converging internationally. MiCA's e-money token regime established a comprehensive framework in the EU effective June 2024. The US GENIUS Act (passed by the Senate in July 2025) would create a federal regulatory framework for payment stablecoins. The UK's cryptoasset regulations (passed February 2026) bring stablecoins within the FCA's regulatory remit. This regulatory convergence is likely to accelerate the integration of regulated stablecoins into the payment system, while marginalising unregulated or algorithmic stablecoins (the Terra/LUNA experience having demonstrated the catastrophic consequences of the latter).

The spot Bitcoin and Ether ETFs have fundamentally changed the accessibility of crypto exposure for institutional investors. By providing a regulated, exchange-traded vehicle with established custody, liquidity, and tax reporting, the ETFs have removed the operational, regulatory, and fiduciary barriers that had excluded most institutional portfolios from direct crypto investment. The \$51.3 billion in cumulative Bitcoin ETF inflows and \$11.0 billion in Ether ETF inflows as of July 2026 represent capital that, in the absence of ETFs, would largely have remained outside crypto markets.

Beyond ETFs, institutional adoption is visible in custody infrastructure (BNY Mellon, State Street), prime brokerage (Coinbase Prime, FalconX), and derivatives markets (CME Bitcoin and Ether futures, options on futures). The CME's crypto derivatives complex regularly processes more volume than spot exchanges, reflecting institutional preference for regulated, centrally cleared markets. The approval of options on spot Bitcoin ETFs in October 2024 further deepened the institutional derivatives infrastructure.

8.1 Tokenisation of Real-World Assets

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9. Research Gaps and Opportunities

The convergence of new regulatory frameworks, standardised data, and unresolved theoretical questions creates fertile ground for accounting, finance, and governance research. The following areas represent particularly promising avenues for scholars working in ABS 4/4* and ABDC A* journals.

The adoption of FASB ASU 2023-08 creates a natural experiment setting. Key research questions include: Does fair value measurement affect the value relevance of crypto holdings relative to the impairment model? How does earnings volatility from fair-value-accounted crypto holdings affect firm valuation and cost of capital? Do firms with fair-value-accounted crypto holdings experience different market reactions to crypto price movements compared to firms reporting under IFRS? Anderson et al. (2026, JAR) have established the pre-ASU baseline; the post-ASU period is now generating the data needed for comparative analysis.

The IFRS-GAAP divergence in crypto accounting provides a cross-jurisdictional research setting. Firms listed on both US and EU exchanges, or firms with peers reporting under different frameworks, allow researchers to isolate the effect of accounting measurement on market outcomes. This divergence also creates opportunities for research on the real effects of accounting standards: does the GAAP treatment (fair value through net income) encourage or discourage corporate crypto holdings relative to the IFRS treatment (cost with optional OCI revaluation)?

The Strategy case raises fundamental questions about corporate treasury management. Is Bitcoin held as a cash substitute, an investment, or a strategic asset? How does the market price firms whose equity value is substantially driven by crypto holdings rather than operating earnings? The literature on corporate cash holdings (Opler et al., 1999; Bates, Kahle, and Stulz, 2009) provides a framework for analysing treasury composition decisions, but crypto introduces a dimension absent from traditional cash management: extreme volatility, fair value accounting through net income, and an evolving regulatory environment.

Gryglewicz, Mayer, and Morellec (2021, JFE) provide a theoretical framework for token financing decisions, showing that token financing is generally preferred to equity for digital platforms unless the platform expects strong cash flows. Chod and Lyandres (2021, MS) compare token and equity financing under conditions of agency conflict and information

asymmetry. These theoretical contributions have yet to be tested empirically with the comprehensive post-ICO, post-ETF data now available.

The staggered adoption of crypto regulation across jurisdictions (MiCA in the EU, ETF approvals in the US, UK cryptoasset regulations in 2026, Australia's Digital Assets Framework in 2026) creates a panel of regulatory events suitable for difference-in-differences analysis. Researchers can examine whether regulatory adoption affects crypto market quality (liquidity, volatility, wash trading), firm entry and exit, the cost of capital for crypto-exposed firms, and the geographic distribution of crypto activity.

The SEC's approval of spot Bitcoin and Ether ETFs is a particularly clean setting. The event date is well defined, the treatment group (assets gaining ETF access) is clearly distinguished from the control group (assets without ETF access), and the outcome variables (price, volume, volatility, liquidity) are high-frequency and observable. Augustin, Rubtsov, and Shin (2023, MS) provide a methodological precedent in their study of Bitcoin futures introduction effects.

Decentralised finance raises questions that span accounting, finance, and information systems. Park (2023, MS) identifies conceptual flaws in automated market makers, including arbitrage losses for liquidity providers and vulnerability to sandwich attacks. Bourveau, Brendel, and Schoenfeld (2024, RAST) provide the first systematic evidence on DeFi smart contract audits. Key open questions include: How do DeFi protocols' governance tokens affect protocol outcomes? Does the absence of a central intermediary change the nature of financial intermediation risk? How should DeFi protocol assets be accounted for by protocol treasuries and token holders?

Stablecoins, as the fastest-growing segment of the crypto market, present their own research opportunities. The shift from unregulated, opaque issuers (Tether) to regulated, transparent issuers (USDC, PYUSD) provides variation in transparency and governance that researchers can exploit. The integration of stablecoins into traditional payment rails (Visa, Mastercard, Stripe) creates settings for studying the competitive dynamics between traditional and crypto-native payment infrastructure.

The audit profession faces both opportunities and challenges from blockchain. Continuous auditing of on-chain transactions is technically feasible but requires new competencies in cryptographic verification, smart contract analysis, and blockchain data extraction. The PCAOB and IAASB have not yet issued guidance on blockchain attestation engagements. Research on the demand for, supply of, and market response to blockchain audit and assurance services is in its infancy.

Proof of reserves, while technically limited, represents a new form of assurance engagement that falls between traditional financial statement audits and system-and-organisation-controls (SOC) reports. The market for PoR services is growing, but the standards governing PoR engagements are not established. Research on the credibility, information content, and limitations of PoR reports would contribute to both the academic literature and the policy debate.

9.1 Accounting for Crypto Assets

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9.5 Audit and Assurance of Blockchain-Based Systems

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10. Conclusion

Blockchain and cryptocurrency have followed an arc from cryptographic experiment to institutional asset class over a period of approximately 17 years. The journey has been marked by

technological innovation (smart contracts, proof-of-stake, layer-2 scaling), market extremes (four boom-bust cycles, culminating in a \$4.31 trillion peak in October 2025), spectacular failures (Mt. Gox, Terra/LUNA, FTX), and the gradual, uneven construction of a regulatory and accounting architecture that was entirely absent when Nakamoto posted the Bitcoin white paper in 2008.

Several structural patterns are now clear. First, the technology has proven more resilient than the institutions built on top of it. The 2022 failures were failures of centralised intermediaries, not of blockchain protocols. Bitcoin and Ethereum continued to process transactions through the worst of the crisis, validating the technological premise of distributed, censorship-resistant ledgers while exposing the governance and risk management failures of the exchanges, lenders, and hedge funds that had grown up around them. Second, the corporate adoption landscape is dominated by a single outlier: Strategy's \$63.7 billion Bitcoin treasury represents an order of magnitude more corporate crypto exposure than any other public company. Whether Strategy is a pioneer that other firms will eventually follow or an idiosyncratic case that reflects the preferences and convictions of a single CEO remains an open question.

Third, regulation is converging but remains fragmented. MiCA provides a comprehensive framework for the EU's 27 member states. The UK, Australia, Japan, Singapore, and Hong Kong have each established distinct regulatory architectures. The US approach remains fragmented across federal agencies and states, though the ETF approvals and the GENIUS Act signal movement toward greater coherence. The FATF Travel Rule and the OECD Crypto-Asset Reporting Framework are creating global standards for AML and tax information exchange, but implementation is uneven. Fourth, the accounting treatment of crypto assets has undergone its most significant change with FASB ASU 2023-08, which replaces the impairment-only model with fair value measurement. The effect of this change on corporate treasury behaviour, financial reporting quality, and market pricing is only beginning to be studied.

For accounting and finance scholars, the post-ETF, post-MiCA, post-ASU 2023-08 period provides a rich institutional setting. The key questions are no longer whether blockchain and crypto matter for business (the evidence is clear that they do) but how they matter: through what channels, with what effects on firms and markets, under what regulatory and accounting frameworks, and with what consequences for investors, auditors, and the broader financial system. The research opportunities identified in this brief are tractable with existing data and methods, and they speak to questions of both academic significance and practical relevance for the many firms, investors, and regulators now navigating a financial landscape in which blockchain-based assets are an established, if still volatile, presence.

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